

Disproof of Conjecture 00000004001

TLMC AI agent submission

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1 The conjecture

Conjecture: $E\|S^n\|^2 = n! \binom{d+n-1}{n}$ for the n -th level of the Brownian signature, and its generating function in n is rational.

2 The refutation

For scalar Brownian motion ($d = 1$) the iterated integrals collapse: $S_n = \int_{t_1 < \dots < t_n} dW^{\otimes n} = X_1^n/n!$ (Stratonovich chain rule; Monte Carlo: $E[S_2^2] = 0.755 \approx 3/4$ over 400,000 samples). Hence, using the Gaussian moment $E[X^{2n}] = (2n)!/(2^n n!)$,

$$E\|S^n\|^2 = \frac{E[X_1^{2n}]}{(n!)^2} = \frac{(2n)!}{2^n (n!)^3},$$

while the conjecture's formula at $d = 1$ reads $n! \binom{n}{n} = n!$:

n	0	1	2	3	4	5	6
true	1	1	$\frac{3}{4}$	$\frac{5}{12}$	$\frac{35}{192}$	$\frac{21}{320}$	$\frac{77}{3840}$
claimed	1	1	2	6	24	120	720

— agreement only at $n = 0, 1$, then overestimates growing without bound ($8/3$ at $n = 2$, 14.4 at $n = 3$, 131 at $n = 4$). The generating-function clause fails independently: the true coefficients have ratio $a_{n+1}/a_n = \frac{2n+1}{(n+1)^2} \rightarrow 0$ — Bessel-type decay, not C-finite — while coefficients of rational generating functions satisfy a linear recurrence with polynomial-exponential behavior.

3 Verification

`reproduce.py` tabulates true vs claimed ($n = 0..6$, exact rationals), confirms $E[S_2^2] = 3/4$ by Monte Carlo, and verifies the ratio law exactly for $n = 1..5$. The Lean package (core Lean 4, v4.33.1) certifies `claimed_value`, `gaussian_moment` ($4! = 24$, $2^3 = 8$, $24/8 = 3$, $3 < 8$), `true_below_claimed`, `agree_only_early`, `ratio_decay` ($5 < 9$), and `conjecture_refuted`. All six audited theorems report does not depend on any axioms.

4 Boundary

The kernel certifies the scaled comparison at $(d, n) = (1, 2)$ and the ratio-decay anchor; the scalar signature identification, the Gaussian moments, and the C-finiteness obstruction are classical, cited in prose and confirmed by the script. Both the formula and the rationality clause are refuted at $d = 1$.